**Course:** Linear Algebra **Instructor:** Sajal Rana
**Homework #1**

## Exercise 1.1

### Question 1.

(a) Linear

(b) Non Linear, contains a product of variables ($x_1x_3$)

(c) Linear, can be re-written as $x_1 + 7x_2 - 3x_3 = 0$

(d) Non Linear, $x_1$ has power other than 1

(e) Non Linear, $x_1$ has power other than 1

(f) Linear

### Question 2.

(a) Linear, $\sin k$ is a constant

(b) Linear, $k$ and $\frac{1}{k}$ are constant coefficients

(c) Linear, $2^k$ is a constant coefficient

### Question 3.

(a) Let $y = t$, and then solving for $x$

$7x - 5t = 3$

$7x = 5t + 3$

$x = \frac{5t}{7} + \frac{3}{7}$

So, we get the solution as: $x = \frac{5t}{7} + \frac{3}{7}, y = t$

(b) Let $x_2 = s, x_3, t$, and then solving for $x1$

$3x_1 - 5s + 4t = 7$

$x_1 = \frac{5s}{3} - \frac{4t}{3} + \frac{7}{3}$

So, we get the solution as: $x_1 = \frac{5s}{3} - \frac{4t}{3} + \frac{7}{3}, x_2 = s, x_3 = t$

(c) Let $x_2 = s, x_3 = t, x_4 = u$, and then solving for $x_1$:

$x_1 = \frac{s}{4} - \frac{5t}{8} + \frac{3u}{4} - \frac{1}{8}$

So, we get the solution as: $x_1 = \frac{s}{4} - \frac{5t}{8} + \frac{3u}{4} - \frac{1}{8}, x_2 = s, x_3 = t, x_4 = u$

(d) Let $w = r, x = s, y = t, z = u$, and then solving for $v$:

$3v - 8r + 2s - t + 4u = 0$

$3v = 8r - 2s + t - 4u$

$v = \frac{8r}{3} - \frac{2s}{3} + \frac{t}{3} - \frac{4u}{3}$

So, we get the solution as: $v = \frac{8r}{3} - \frac{2s}{3} + \frac{t}{3} - \frac{4u}{3}, w = r, x = s, y = t, z = u$

**Question 4.**

(c)

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & -1 & 1 & 1 \\ 0 & 3 & 1 & 0 & -1 & 2 \\ 0 & 0 & 1 & 7 & 0 & 1 \end{array}\right]$$

(d)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

**Question 5.**

(c)

$$\begin{aligned} 7x_1 + 2x_2 + x_3 - 3x_4 &= 5 \\ x_1 + 2x_2 + 4x_3 &= 1 \end{aligned}$$

**Question 7.** Since the curve $y = ax^2 + bx + c$ passes through $(x_1, y_1)$, $(x_2, y_2)$, and $(x_3, y_3)$, substituting these points into the given equation of the curve yields:

$$\begin{aligned} ax_1^2 + bx_1 + c &= y_1 \\ ax_2^2 + bx_2 + c &= y_2 \\ ax_3^2 + bx_3 + c &= y_3 \end{aligned}$$

Treating $a, b$, and $c$ as the unknowns, the corresponding augmented matrix for this system of linear equations is:

$$\begin{bmatrix} x_1^2 & x_1 & 1 & y_1 \\ x_2^2 & x_2 & 1 & y_2 \\ x_3^2 & x_3 & 1 & y_3 \end{bmatrix}$$

**Question 8.**

Rewriting the equations as:

$$x_1 = -kx_2 + c \tag{1}$$

$$x_1 = -lx_2 + d \tag{2}$$

For $x_2 = 0$, (1) gives $x_1 = c$. The pair $(c, 0)$ also satisfies (2):

$$c = 0 + d$$

$$\text{or } c = d$$

Therefore

$$x_1 + kx_2 = x_1 + lx_2$$

$$\text{or } k = l$$

Since $k = l$ and $c = d$, the equations are identical.

## Exercise 1.3

**Question 1.**

$$A\ (4 \times 5), \qquad B\ (4 \times 5), \qquad C\ (5 \times 2), \qquad D\ (4 \times 2), \qquad E\ (5 \times 4).$$

(a) $BA$

$B$ is $4 \times 5$ and $A$ is $4 \times 5$. For $BA$ to be defined, the number of columns of $B$ must equal the number of rows of $A$.

$$BA = (4 \times 5)(4 \times 5)$$

Since $5 \neq 4$, the inner dimensions do not match. Therefore, $BA$ is not defined.

(b) $AC + D$

$A$ is $4 \times 5$ and $C$ is $5 \times 2$, so

$$AC = (4 \times 5)(5 \times 2) = 4 \times 2.$$

Since $D$ is also $4 \times 2$, the addition is defined.

$$AC + D = (4 \times 2) + (4 \times 2) = 4 \times 2.$$

Therefore, $AC + D$ is defined and the resulting matrix has size $4 \times 2$.

(c) $AE + B$

$A$ is $4 \times 5$ and $E$ is $5 \times 4$, so

$$AE = (4 \times 5)(5 \times 4) = 4 \times 4.$$

However, $B$ is $4 \times 5$. Therefore,

$$(4 \times 4) + (4 \times 5)$$

is not defined. Hence, $AE + B$ is not defined.

(d) $AB + B$

$A$ is $4 \times 5$ and $B$ is $4 \times 5$. For $AB$ to be defined,

$$AB = (4 \times 5)(4 \times 5).$$

Since $5 \neq 4$, the inner dimensions do not match. Therefore, $AB$ is not defined, and hence $AB + B$ is not defined.

(e) $E(A + B)$

Since $A$ and $B$ are both $4 \times 5$ matrices,

$$A + B = (4 \times 5) + (4 \times 5) = 4 \times 5.$$

Since $E$ is $5 \times 4$,

$$E(A + B) = (5 \times 4)(4 \times 5) = 5 \times 5.$$

Therefore, $E(A + B)$ is defined and the resulting matrix has size $5 \times 5$.

(f) $E(AC)$

First,

$$AC = (4 \times 5)(5 \times 2) = 4 \times 2.$$

Since $E$ is $5 \times 4$,

$$E(AC) = (5 \times 4)(4 \times 2) = 5 \times 2.$$

Therefore, $E(AC)$ is defined and the resulting matrix has size $5 \times 2$.

(g) $E^T A$

Since $E$ is $5 \times 4$, its transpose is

$$E^T = 4 \times 5.$$

Therefore,

$$E^T A = (4 \times 5)(4 \times 5).$$

Since $5 \neq 4$, the inner dimensions do not match. Therefore, $E^T A$ is not defined.

(h) $(A^T + E)D$

Since $A$ is $4 \times 5$, its transpose is

$$A^T = 5 \times 4.$$

Since $E$ is also $5 \times 4$,

$$A^T + E = (5 \times 4) + (5 \times 4) = 5 \times 4.$$

Since $D$ is $4 \times 2$,

$$(A^T + E)D = (5 \times 4)(4 \times 2) = 5 \times 2.$$

Therefore, $(A^T + E)D$ is defined and the resulting matrix has size $5 \times 2$.

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**Question 7.**

Using the method of Example 7 to find the following.

(a) **The first row of $AB$, i.e. $A_1B$.**

We multiply the first row of $A$ by $B$:

$$\begin{bmatrix}3 & -2 & 7\end{bmatrix}\begin{bmatrix}6 & -2 & 4\\ 0 & 1 & 3\\ 7 & 7 & 5\end{bmatrix} = \begin{bmatrix}18+0+49 & -6-2+49 & 12-6+35\end{bmatrix} = \begin{bmatrix}67 & 41 & 41\end{bmatrix}.$$

(b) **The third row of $AB$, i.e. $A_3B$.**

We multiply the third row of $A$ by $B$:

$$\begin{bmatrix}0 & 4 & 9\end{bmatrix}\begin{bmatrix}6 & -2 & 4\\ 0 & 1 & 3\\ 7 & 7 & 5\end{bmatrix} = \begin{bmatrix}0+0+63 & 0+4+63 & 0+12+45\end{bmatrix} = \begin{bmatrix}63 & 67 & 57\end{bmatrix}.$$

(c) **The second column of $AB$, i.e. $AB_2$.**

We multiply $A$ by the second column of $B$:

$$\begin{bmatrix}3 & -2 & 7\\ 6 & 5 & 4\\ 0 & 4 & 9\end{bmatrix}\begin{bmatrix}-2\\ 1\\ 7\end{bmatrix} = \begin{bmatrix}-6-2+49\\ -12+5+28\\ 0+4+63\end{bmatrix} = \begin{bmatrix}41\\ 21\\ 67\end{bmatrix}.$$

(d) **The first column of $BA$, i.e. $BA_1$.**

We multiply $B$ by the first column of $A$:

$$\begin{bmatrix}6 & -2 & 4\\ 0 & 1 & 3\\ 7 & 7 & 5\end{bmatrix}\begin{bmatrix}3\\ 6\\ 0\end{bmatrix}$$

$$\begin{bmatrix}18-12+0\\ 0+6+0\\ 21+42+0\end{bmatrix} = \begin{bmatrix}6\\ 6\\ 63\end{bmatrix}.$$

(e) **The third row of $AA$, i.e. $A_3A$.**

We multiply the third row of $A$ by $A$:

$$\begin{bmatrix}0 & 4 & 9\end{bmatrix}\begin{bmatrix}3 & -2 & 7\\ 6 & 5 & 4\\ 0 & 4 & 9\end{bmatrix} = \begin{bmatrix}0+24+0 & 0+20+36 & 0+16+81\end{bmatrix} = \begin{bmatrix}24 & 56 & 97\end{bmatrix}.$$

(f) **The third column of $AA$, i.e. $AA_3$.**

We multiply $A$ by the third column of $A$:

$$\begin{bmatrix}3 & -2 & 7\\ 6 & 5 & 4\\ 0 & 4 & 9\end{bmatrix}\begin{bmatrix}7\\ 4\\ 9\end{bmatrix} = \begin{bmatrix}21-8+63\\ 42+20+36\\ 0+16+81\end{bmatrix} = \begin{bmatrix}76\\ 98\\ 97\end{bmatrix}.$$

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**Question 8.**

$$A = \begin{bmatrix}3 & -2 & 7\\ 6 & 5 & 4\\ 0 & 4 & 9\end{bmatrix}, \quad B = \begin{bmatrix}6 & -2 & 4\\ 0 & 1 & 3\\ 7 & 7 & 5\end{bmatrix}$$

(a)

To find the column matrices of $AB$, we multiply the columns of $A$ by the scalar entries from the corresponding column of $B$ and sum them :

- First column of $AB$:

$$6\begin{bmatrix}3\\6\\0\end{bmatrix} + 0\begin{bmatrix}-2\\5\\4\end{bmatrix} + 7\begin{bmatrix}7\\4\\9\end{bmatrix} = \begin{bmatrix}18\\36\\0\end{bmatrix} + \begin{bmatrix}0\\0\\0\end{bmatrix} + \begin{bmatrix}49\\28\\63\end{bmatrix} = \begin{bmatrix}67\\64\\63\end{bmatrix}$$

:

$$\begin{bmatrix}67\\64\\63\end{bmatrix} = 6\begin{bmatrix}3\\6\\0\end{bmatrix} + 0\begin{bmatrix}-2\\5\\4\end{bmatrix} + 7\begin{bmatrix}7\\4\\9\end{bmatrix}$$

- Second column of $AB$:

$$-2\begin{bmatrix}3\\6\\0\end{bmatrix} + 1\begin{bmatrix}-2\\5\\4\end{bmatrix} + 7\begin{bmatrix}7\\4\\9\end{bmatrix} = \begin{bmatrix}-6\\-12\\0\end{bmatrix} + \begin{bmatrix}-2\\5\\4\end{bmatrix} + \begin{bmatrix}49\\28\\63\end{bmatrix} = \begin{bmatrix}41\\21\\67\end{bmatrix}$$

:

$$\begin{bmatrix}41\\21\\67\end{bmatrix} = -2\begin{bmatrix}3\\6\\0\end{bmatrix} + 1\begin{bmatrix}-2\\5\\4\end{bmatrix} + 7\begin{bmatrix}7\\4\\9\end{bmatrix}$$

- Third column of $AB$:

$$4\begin{bmatrix}3\\6\\0\end{bmatrix} + 3\begin{bmatrix}-2\\5\\4\end{bmatrix} + 5\begin{bmatrix}7\\4\\9\end{bmatrix} = \begin{bmatrix}12\\24\\0\end{bmatrix} + \begin{bmatrix}-6\\15\\12\end{bmatrix} + \begin{bmatrix}35\\20\\45\end{bmatrix} = \begin{bmatrix}41\\59\\57\end{bmatrix}$$

:

$$\begin{bmatrix}41\\59\\57\end{bmatrix} = 4\begin{bmatrix}3\\6\\0\end{bmatrix} + 3\begin{bmatrix}-2\\5\\4\end{bmatrix} + 5\begin{bmatrix}7\\4\\9\end{bmatrix}$$

(b)

Similarly, to find the column matrices of $BA$, we treat $B$ as the leading matrix. We multiply the columns of $B$ by the scalar entries from the corresponding column of $A$ and sum them :

- First column of $BA$:

$$3\begin{bmatrix}6\\0\\7\end{bmatrix} + 6\begin{bmatrix}-2\\1\\7\end{bmatrix} + 0\begin{bmatrix}4\\3\\5\end{bmatrix} = \begin{bmatrix}18\\0\\21\end{bmatrix} + \begin{bmatrix}-12\\6\\42\end{bmatrix} + \begin{bmatrix}0\\0\\0\end{bmatrix} = \begin{bmatrix}6\\6\\63\end{bmatrix}$$

:

$$\begin{bmatrix}6\\6\\63\end{bmatrix} = 3\begin{bmatrix}6\\0\\7\end{bmatrix} + 6\begin{bmatrix}-2\\1\\7\end{bmatrix} + 0\begin{bmatrix}4\\3\\5\end{bmatrix}$$

- Second column of $BA$:

$$-2\begin{bmatrix}6\\0\\7\end{bmatrix} + 5\begin{bmatrix}-2\\1\\7\end{bmatrix} + 4\begin{bmatrix}4\\3\\5\end{bmatrix} = \begin{bmatrix}-12\\0\\-14\end{bmatrix} + \begin{bmatrix}-10\\5\\35\end{bmatrix} + \begin{bmatrix}16\\12\\20\end{bmatrix} = \begin{bmatrix}-6\\17\\41\end{bmatrix}$$

:

$$\begin{bmatrix}-6\\17\\41\end{bmatrix} = -2\begin{bmatrix}6\\0\\7\end{bmatrix} + 5\begin{bmatrix}-2\\1\\7\end{bmatrix} + 4\begin{bmatrix}4\\3\\5\end{bmatrix}$$

- Third column of $BA$:

$$7\begin{bmatrix}6\\0\\7\end{bmatrix} + 4\begin{bmatrix}-2\\1\\7\end{bmatrix} + 9\begin{bmatrix}4\\3\\5\end{bmatrix} = \begin{bmatrix}42\\0\\49\end{bmatrix} + \begin{bmatrix}-8\\4\\28\end{bmatrix} + \begin{bmatrix}36\\27\\45\end{bmatrix} = \begin{bmatrix}70\\31\\122\end{bmatrix}$$

:

$$\begin{bmatrix}70\\31\\122\end{bmatrix} = 7\begin{bmatrix}6\\0\\7\end{bmatrix} + 4\begin{bmatrix}-2\\1\\7\end{bmatrix} + 9\begin{bmatrix}4\\3\\5\end{bmatrix}$$

**Question 12.**

(Note that we will always be using general matrix sizes and not any specific size to prove/show)

(a) Let the size of matrix $A$ be $m \times n$.

For the product $AB$ to be defined, the number of rows in matrix $B$ must equal the number of columns in $A$, meaning $B$ must have $n$ rows.

Let the size of matrix $B$ be $n \times p$. The resulting matrix $AB$ will have size $m \times p$.

For the product $BA$ to be defined, the number of columns of $B$ (which is $p$) must equal the number of rows of $A$ (which is $m$) . Therefore, $p = m$.

This means $B$ must be of size $n \times m$. Consequently, the product $AB$ has size $m \times m$, which is a square matrix. Similarly, the product $BA$ will have size $n \times n$ (since $B$ is $n \times m$ and $A$ is $m \times n$), making it a square matrix as well .

(b)

We are given that $A$ is an $m \times n$ matrix. Let the unknown size of matrix $B$ be $r \times s$.
For the product $BA$ to be defined, the number of columns of $B$ $(s)$ must equal the number of rows of $A$ $(m)$ . Thus, $s = m$, meaning $B$ is an $r \times m$ matrix, and the resulting product $BA$ is an $r \times n$ matrix .
Next, for the product $A(BA)$ to be defined, the number of columns of $A$ $(n)$ must match the number of rows of $BA$ $(r)$ . Therefore, $r = n$. Since $r = n$ and $s = m$, the matrix $B$ must be an $n \times m$ matrix .

**Question 13.**

(a)

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 9 & -1 & 1 \\ 1 & 5 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ -1 \\ 0 \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} 4 & 0 & -3 & 1 \\ 5 & 1 & 0 & -8 \\ 2 & -5 & 9 & -1 \\ 0 & 3 & -1 & 7 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

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**Question 17.**

**Part (a)**

The column partitions of $A$ must match the row partitions of $B$. Specifically, the number of columns in $A_{11}$ must equal the number of rows in $B_{11}$ for the product $A_{11}B_{11}$ to be defined. Here, the first block column of $A$ has 3 columns, but the first block row of $B$ has 2 rows. Since $3 \neq 2$, the submatrix products like $A_{11}B_{11}$ are undefined. Therefore, block multiplication **cannot** be used for these partitions.

**Part (b)**

Matrix $A$ is partitioned as $A = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix}$, where $A_{11}$ has dimensions $2 \times 4$ and $A_{21}$ has dimensions $1 \times 4$. Matrix $B$ is partitioned as $B = \begin{bmatrix} B_{11} & B_{12} & B_{13} \end{bmatrix}$, where $B_{11}$, $B_{12}$, and $B_{13}$ each have dimensions $4 \times 1$.

Since all blocks in $A$ have 4 columns and all blocks in $B$ have 4 rows, the inner dimensions match for every individual block product. Therefore, block multiplication **can** be used.

The block product structure is defined as:

$$AB = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} & A_{11}B_{12} & A_{11}B_{13} \\ A_{21}B_{11} & A_{21}B_{12} & A_{21}B_{13} \end{bmatrix}$$

Computing the individual block products:

$$A_{11}B_{11} = \begin{bmatrix} -1 & 2 & 1 & 5 \\ 0 & -3 & 4 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 7 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 37 \end{bmatrix}$$

$$A_{11}B_{12} = \begin{bmatrix} -1 & 2 & 1 & 5 \\ 0 & -3 & 4 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 23 \\ -13 \end{bmatrix}$$

$$A_{11}B_{13} = \begin{bmatrix} -1 & 2 & 1 & 5 \\ 0 & -3 & 4 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \\ -3 \end{bmatrix} = \begin{bmatrix} -10 \\ 8 \end{bmatrix}$$

$$A_{21}B_{11} = \begin{bmatrix} 1 & 5 & 6 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 29 \end{bmatrix}$$

$$A_{21}B_{12} = \begin{bmatrix} 1 & 5 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 23 \end{bmatrix}$$

$$A_{21}B_{13} = \begin{bmatrix} 1 & 5 & 6 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 41 \end{bmatrix}$$

Assembling the final product matrix:

$$AB = \begin{bmatrix} -1 & 23 & -10 \\ 37 & -13 & 8 \\ 29 & 23 & 41 \end{bmatrix}$$

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**Question 18.**

**Part (a)**

**Method 1**

Let $A$ be an $m \times n$ matrix and $B$ be an $n \times p$ matrix. Then the product $AB$ is defined and is an $m \times p$ matrix . Suppose that the $i$-th row of matrix $A$ is a zero row, that is,

$$\text{row } i \text{ of } A = \begin{bmatrix} 0 & 0 & \cdots & 0 \end{bmatrix}$$

By the definition of matrix multiplication by rows, the $i$-th row of $AB$ is given by :

$$\text{row } i \text{ of } AB = (\text{row } i \text{ of } A) \times B$$

Since the $i$-th row of $A$ is the zero vector, we have

$$\mathbf{0} \times B = \mathbf{0}$$

Therefore, the $i$-th row of $AB$ is also a zero row. Hence, $AB$ has a row of zeros.

**Method 2**

**Proof.** Let $A$ be an $m \times n$ matrix and $B$ be an $n \times p$ matrix. Since $AB$ is defined, these dimensions are compatible. Assume that the $i$-th row of $A$ is a row of zeros. That is,

$$a_{i1} = a_{i2} = \cdots = a_{in} = 0.$$

We claim that the $i$-th row of the matrix $AB$ is also a row of zeros. Let $(AB)_{ij} = c_{ij}$ be an arbitrary entry in the $i$-th row of $AB$. By the definition of matrix multiplication ,

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Since the $i$-th row of $A$ is a zero row, we have

$$a_{i1} = a_{i2} = \cdots = a_{in} = 0.$$

Therefore,

$$c_{ij} = (0)b_{1j} + (0)b_{2j} + \cdots + (0)b_{nj} = \sum_{k=1}^{n} (0)b_{kj} = 0.$$

Since $j$ was arbitrary, every entry in the $i$-th row of $AB$ is zero. Hence, the $i$-th row of $AB$ is a row of zeros.

Therefore, if $A$ has a row of zeros, then $AB$ also has a row of zeros.

**Method 3**

In general, if $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix :

$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

the entry $(AB)_{ij}$ in row $i$ and column $j$ of $AB$ is given by

$$(AB)_{ij} = c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Let the $i$-th row of $A$ be zero, so we have $a_{i1} = a_{i2} = \cdots = a_{in} = 0$.

$$c_{ij} = (0)b_{1j} + (0)b_{2j} + \cdots + (0)b_{nj} = \sum_{k=1}^{n} (0)b_{kj} = 0.$$

Hence, the $i$-th row of $AB$ is all zeros, i.e., $\sum_{k=1}^{n} a_{ik}b_{kj} = 0$.

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